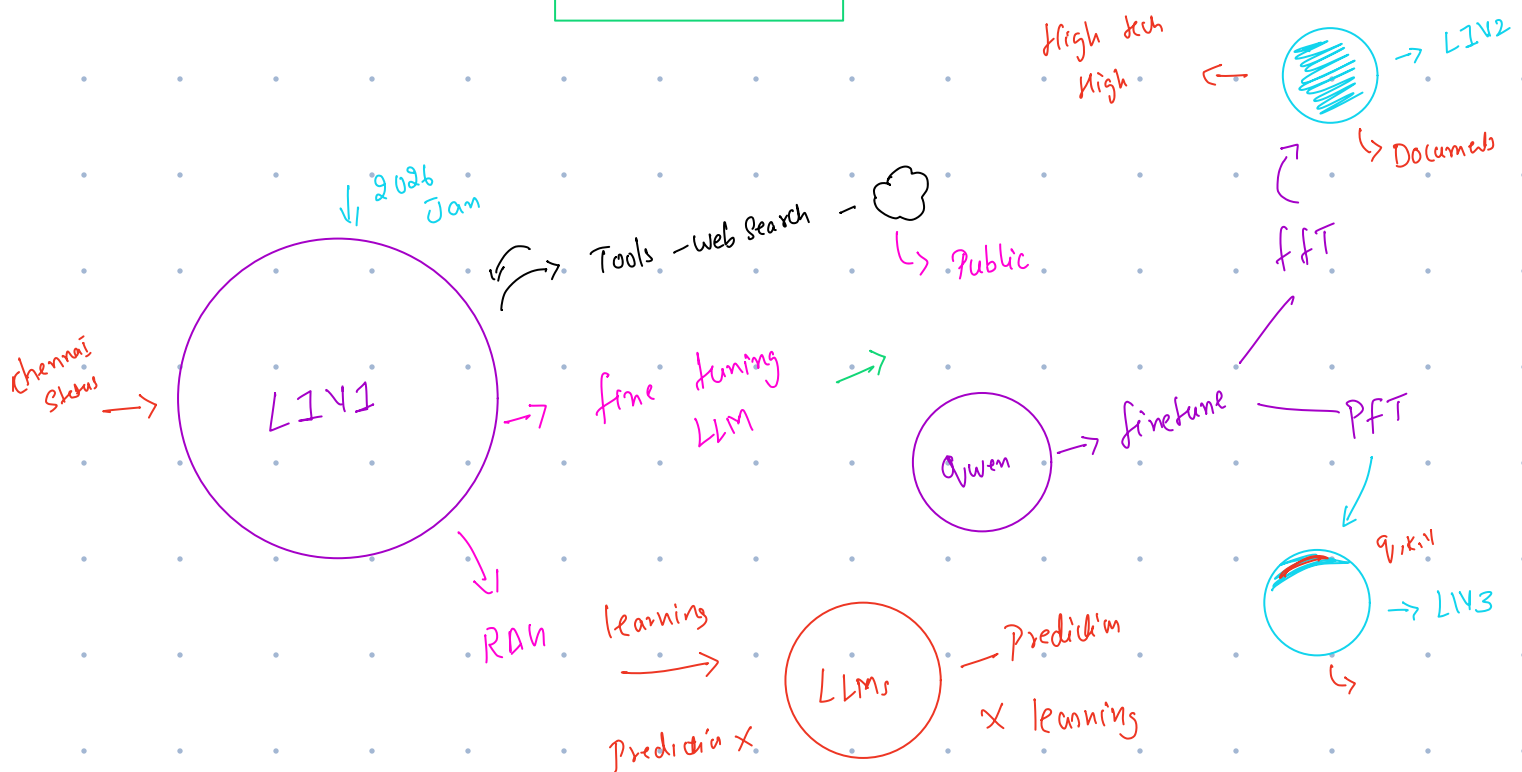
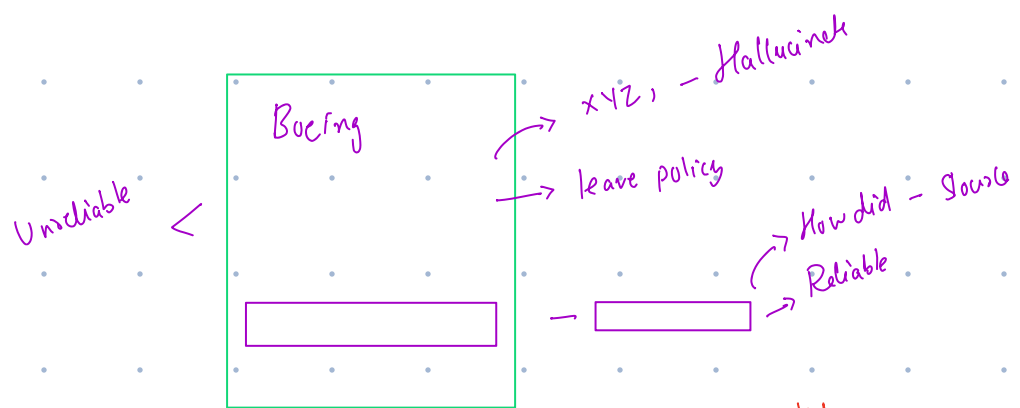
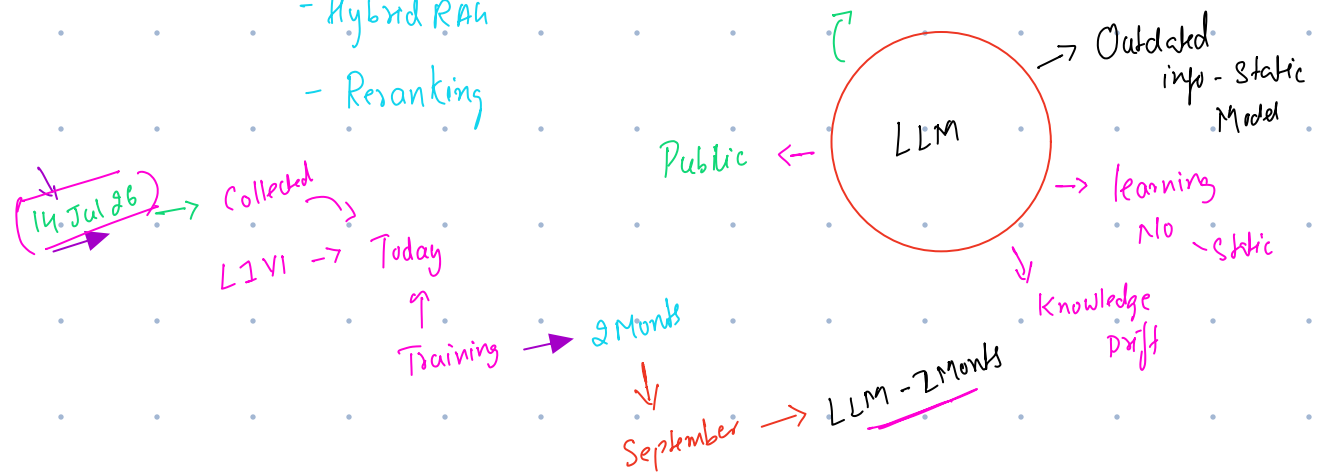
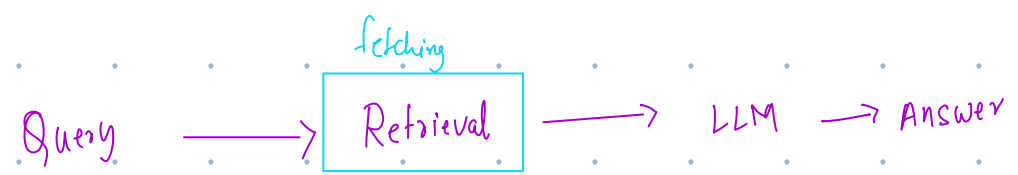


* RAG

- lifecycle
- Tracking
- Hybrid RAN
- Resampling

Need

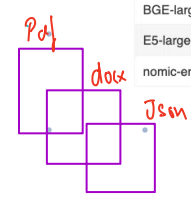




Stage 1 $\rightarrow$

Model	Dim	Context	Notes
OpenAI text-embedding-3-small	1536	8191	Cheap, fast, solid default.
OpenAI text-embedding-3-large	3072	8191	Highest-quality closed model from OpenAI.
Cohere embed-v3	1024	512	Strong multilingual, has query/doc modes.
Voyage-3	1024	32k	Very long context; strong on retrieval benchmarks.
BGE-large (BAAI)	1024	512	Open-source, top MTEB scores.
E5-large-v2	1024	512	Open-source, lightweight, asymmetric.
nomic-embed-text-v1.5	768	8192	Open-source, long-context, Matryoshka.

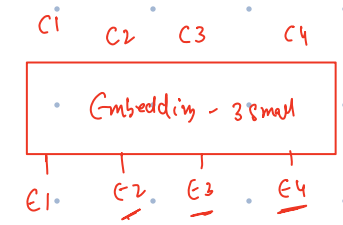
(Retrieved)



Document loaders



Chunking



Embeddings

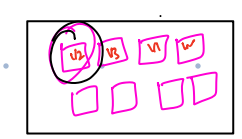
CS = 500
Overlap = 100

Cosine Similarity

$v_1, v_2 \approx 0.90$

$v_2 =$
C = "Text"

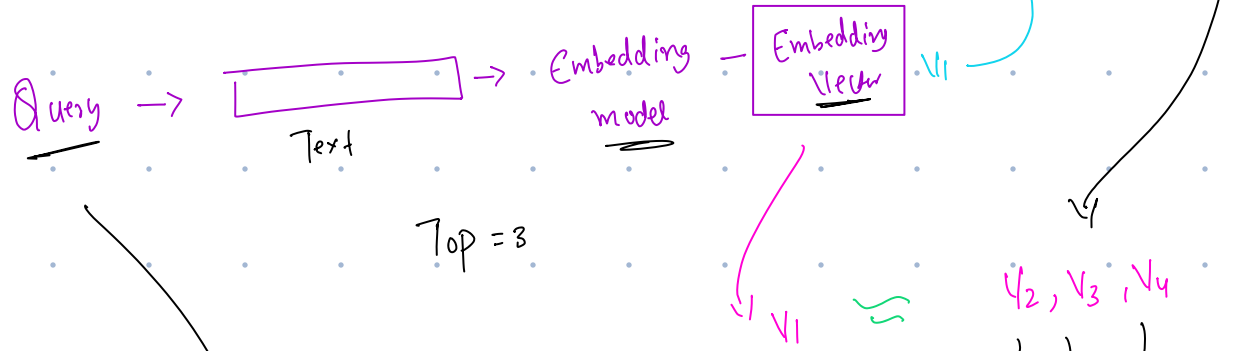
c_2, c_3, c_4



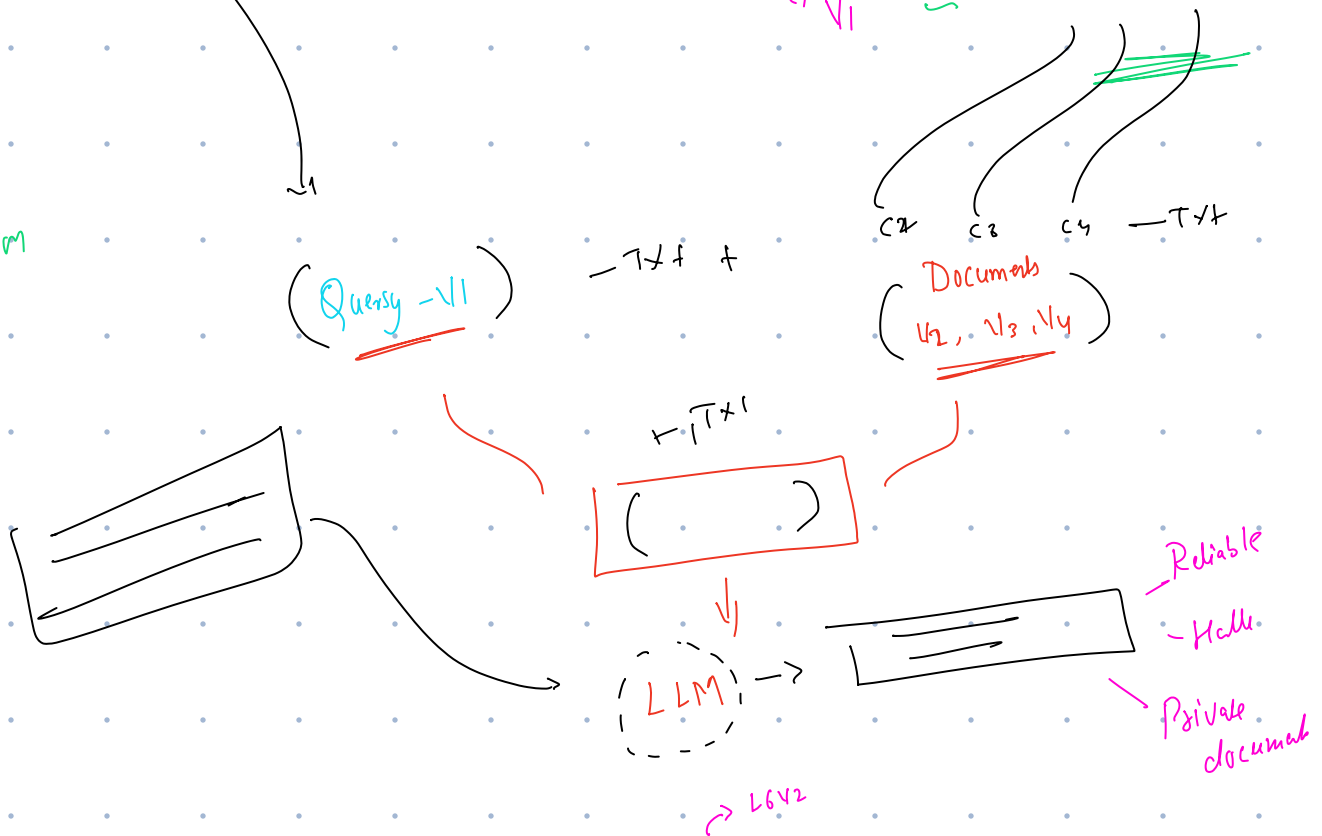
Vector Database

$v_1 \approx v_2 = 0.90$
 $v_1 \approx v_3 = 0.80$
 $v_1 \approx v_4 = 0.85$

Retriever

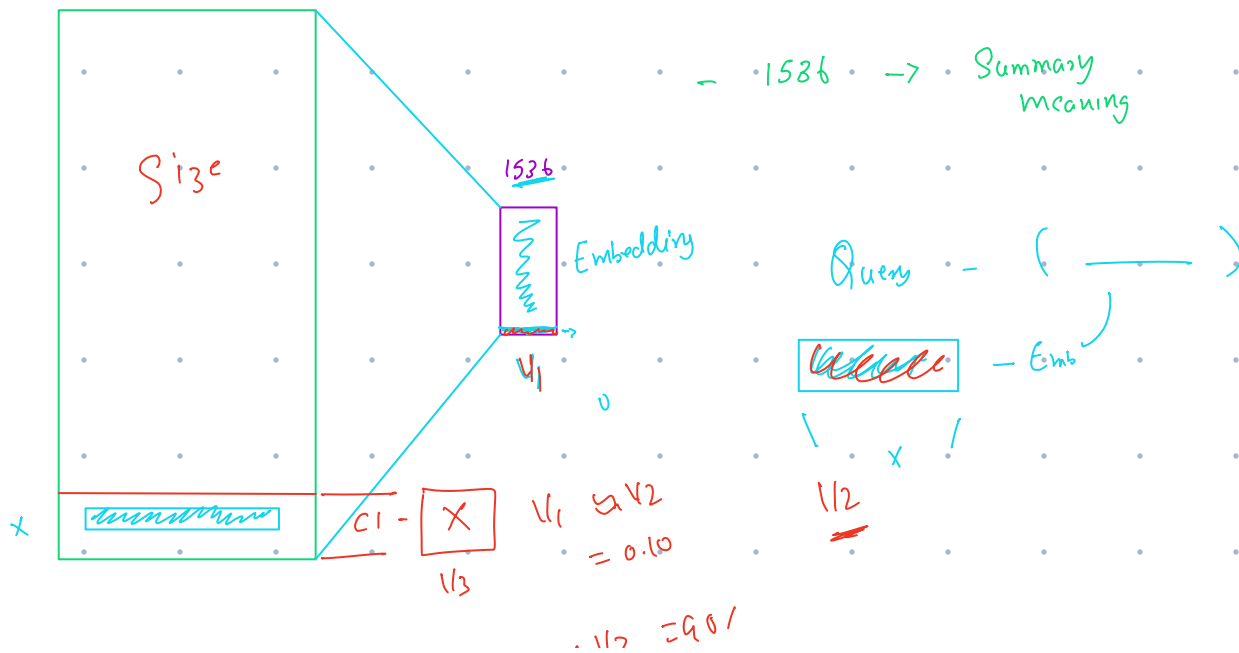


Generation



Sentence $\rightarrow$ Embedding $\rightarrow$ 1536, 384, 768, 512

$\hookrightarrow$ BGE



1/2 2 ">

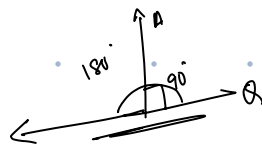
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Use Case	Recommended Chunk Size	Overlap	Strategy
General Q&A over documents	400-600 tokens	50-100 tokens	Recursive Character
Code retrieval	Full functions/classes	0 (function boundary)	Language-aware splitter
Legal / contract analysis	300-500 tokens	100 tokens	Semantic or Structure-aware
Medical literature	200-400 tokens	50 tokens	Semantic Chunking
Customer support FAQ	100-200 tokens	20-50 tokens	Sentence-level
Long research papers	500-800 tokens	100-150 tokens	Semantic or Header-based
Real-time fact retrieval	50-150 tokens (propositions)	0	Proposition-based

Document
loader
($S_1 \rightarrow E$
 $S_2 \rightarrow E$
 S_3)

Index Type	Recall	Speed	Memory	Build Time	Best For
Flat (Exact)	100%	Slow (O(N))	Low	Instant	< 100K vectors
(HNSW)	95-99%	Very fast	High	Moderate	Most production RAG
IVF-Flat	90-98%	Fast	Low-Medium	Moderate	100K-10M vectors
IVF-PQ	85-95%	Very fast	Very low	Slow	10M+ vectors, memory constrained
ScaNN (Google)	98%	Fastest at scale	Medium	Slow	Extreme scale

$\cos \theta = 1$
 $\cos 1 = 0$
 $\cos 180 = -1$



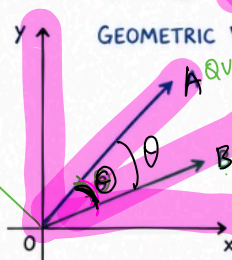
HOW COSINE SIMILARITY WORKS

Measure of similarity between two vectors based on the angle between them, ignoring their magnitude.

INTUITION

- Treat vectors as arrows from the origin.
- The smaller the angle between them, the more similar they are.
- Magnitude (length) doesn't matter—only direction does.

GEOMETRIC VIEW



θ = angle between A and B
 If $\theta = 0^\circ \rightarrow$ identical direction
 If $\theta = 90^\circ \rightarrow$ no similarity
 If $\theta = 180^\circ \rightarrow$ opposite direction

RANGE

1 $\rightarrow$ identical ($\theta = 0^\circ$)
 0 $\rightarrow$ orthogonal / no similarity ($\theta = 90^\circ$)
 -1 $\rightarrow$ opposite ($\theta = 180^\circ$)
 Range: $[-1, 1]$

THE MATH

Given two vectors:

$A = [a_1, a_2, \dots, a_n]$

$B = [b_1, b_2, \dots, b_n]$

Cosine Similarity $(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$

1) Dot Product ($A \cdot B$)

$A \cdot B = \sum_{i=1}^n a_i b_i$

Measures how much the vectors point in the same direction.

2) Magnitude (Length)

$\|A\| = \sqrt{\sum_{i=1}^n a_i^2}$

$\|B\| = \sqrt{\sum_{i=1}^n b_i^2}$

Always a non-negative value.

3) Final Formula

$\cos(\theta) = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n b_i^2}}$

Normalized similarity based only on the angle between vectors.

NUMERICAL EXAMPLE

Let

$A = [1, 2, 3]$

$B = [4, 5, 6]$

Step 1: Dot Product

$A \cdot B = (1 \times 4) + (2 \times 5) + (3 \times 6)$
 $= 4 + 10 + 18 = 32$

Step 2: Magnitudes

$\|A\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} \approx 3.742$

$\|B\| = \sqrt{4^2 + 5^2 + 6^2} = \sqrt{77} \approx 8.775$

Step 3: Cosine Similarity

$\cos(\theta) = \frac{32}{3.742 \times 8.775}$
 $= \frac{32}{32.83} \approx 0.974$

Interpretation: θ is small (close to 0°) $\rightarrow$ vectors point in almost the same direction $\rightarrow$ very similar.

HOW IT WORKS IN PRACTICE (e.g., RAG)

1. Documents



Collection of documents

2. Convert to Vectors

Embedding Model
 Each document $\rightarrow$ high-dimensional vector
 [0.12, -0.45, ..., 0.33]
 [0.31, 0.22, ..., -0.11]
 ...
 [-0.05, 0.78, ..., 0.14]

3. Query

"What are the benefits of exercise?"
 Embedding Model
 [0.28, -0.10, ..., 0.42]
 Query vector

4. Compute Cosine Similarity

Doc 1 [0.12, -0.45, ..., 0.33] $\cos = 0.82$
 Doc 2 [0.31, 0.22, ..., -0.11] $\cos = 0.15$
 ...
 Doc N [-0.05, 0.78, ..., 0.14] $\cos = -0.10$

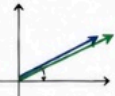
5. Retrieve Top-K

Top similar documents
 1. _____
 2. _____
 3. _____
 ...

Documents with higher cosine similarity to the query are considered more relevant.

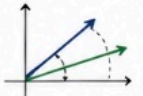
VISUAL SUMMARY

High Similarity ($\cos \approx 1$)



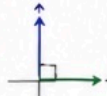
Small angle
Very similar

Medium Similarity ($0 < \cos < 1$)



Moderate angle
Somewhat similar

No Similarity ($\cos \approx 0$)



90° angle
Not similar

Opposite Direction ($\cos \approx -1$)



180° angle
Opposite

KEY TAKEAWAYS

- ✓ Cosine similarity measures the angle between vectors, not their length.
- ✓ Values range from -1 to 1.
- ✓ Widely used in information retrieval, recommendation systems, NLP, and RAG.
- ✓ Fast, simple, and effective in high-dimensional spaces.

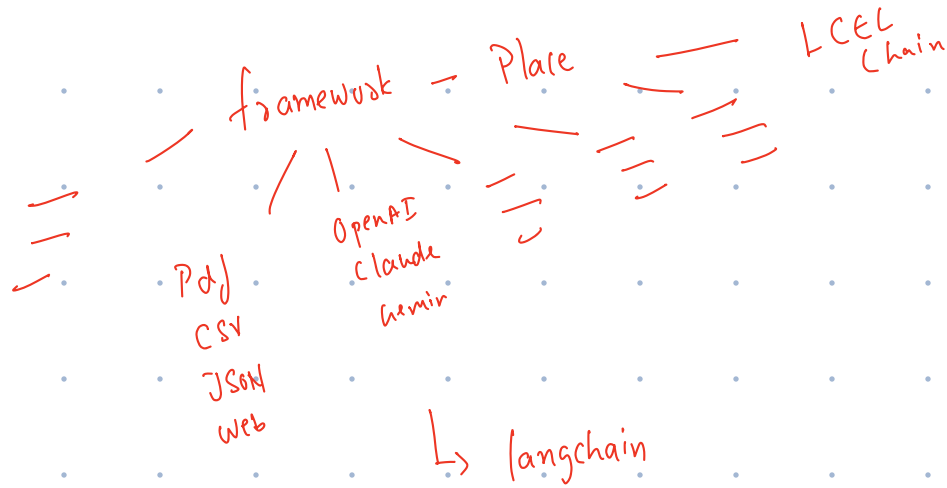
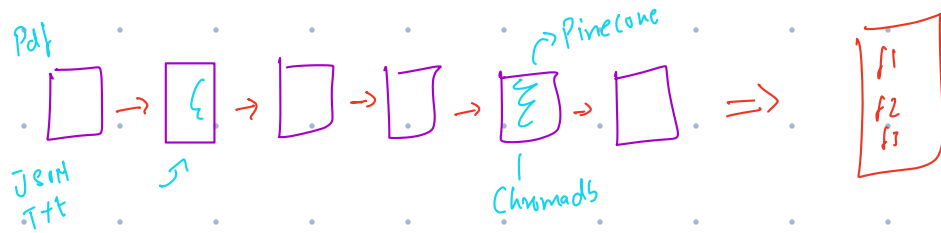
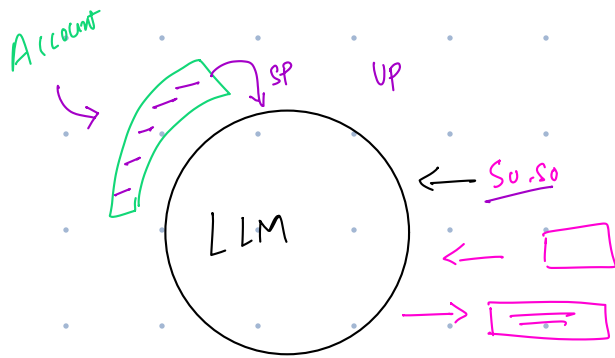
$V_1 - A$

$V_2 - B$

V_3

V_4

$0.974 \approx 1$ = similar
 $0.15 \rightarrow$



framework - GenAI flows

↳ Components — Composable
↳ Integrations

